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Resilient infrastructure planning post-disaster

Building Back Better: Strategies for Resilient Infrastructure Planning and Recovery Post-Disaster

Quick Reference

Key Findings Table

Theme	Key Insights	Supporting Citations
Multi-hazard risk assessment	Dynamic Bayesian Networks, geospatial integration, probabilistic models, and multi-criteria decision analysis	1 2 3
Legislative and policy frameworks	Decentralized recovery authorities, risk-informed land-use, multi-hazard codes, community empowerment	4 5
Community involvement and empowerment	Participatory planning, education, social capital, bottom-up governance, legitimacy, transparency	6 7 8
Technological innovations	AI, IoT, GIS, digital platforms, performance-based risk assessment	9 10 11
Recovery strategies and frameworks	Sustainable recovery, social capital integration, decentralized authorities, PCHR models	12 13 14
Power dynamics and governance	Institutional reforms, fiscal decentralization, transparency, trust-building	8 15
Research gaps	Quantitative evaluation of community interventions, long-term legislative impact, power dynamics integration	16 17 18

Direct Answer

Refer to research papers on:

- Integrated multi-hazard risk assessment methodologies (Dynamic Bayesian Networks, geospatial technologies)
- Legislative models and community empowerment in post-disaster recovery
- Technological innovations (AI, IoT, GIS) in risk-informed resilient infrastructure planning

These topics capture the core aspects of the Build Back Better framework and provide a holistic understanding of planning resilient infrastructure post-disaster.

Study Scope

- **Time Period:** Primarily post-2015, reflecting Sendai Framework and contemporary BBB applications
- **Disciplines:** Disaster risk reduction, civil engineering, urban planning, social sciences, policy studies, information technology
- **Methods:** Meta-analysis, probabilistic modeling, geospatial integration, qualitative case studies, participatory frameworks

Assumptions & Limitations

- Assumes availability of multi-source data for risk assessment and community engagement
- Limitations in empirical evaluation of long-term impacts and scalability across diverse contexts
- Power dynamics and social processes often underexamined in practical recovery literature
- Resource constraints and governance issues may limit implementation in low-income settings

Suggested Further Research

- Quantitative assessment of community education and participatory recovery schemes
- Integration of power dynamics into recovery models
- Empirical evaluation of decentralized legislative models across cultural/economic contexts
- Interdisciplinary studies linking technological innovations with ground-level recovery outcomes

1. Introduction

Context and Purpose

The Build Back Better (BBB) framework has emerged as a cornerstone of post-disaster recovery, emphasizing not only restoration but enhancement of infrastructure resilience, community empowerment, and sustainable development. The increasing frequency and severity of disasters globally have underscored the need for robust, risk-informed planning that leverages both technological innovation and participatory governance. This report synthesizes research on resilient infrastructure planning post-disaster, bridging theoretical foundations with practical insights to inform BBB implementation across diverse contexts [19] [20] [21].

2. Core Principles of Disaster Resilience in Infrastructure Planning

Multi-Hazard Risk Assessment and Risk-Informed Planning

- **Dynamic Bayesian Networks (DBNs):** Enable temporal, multi-hazard risk assessment of interconnected infrastructure, capturing real-time disruptions and cascading effects [1] [2].
- **Geospatial Integration:** Harmonizes hazard maps into unified frameworks, accounting for hazard interdependencies and critical infrastructure exposure [3] [22].
- **Probabilistic and Index-Based Methods:** Combine vulnerability indicators and structural risk to prioritize mitigation actions at regional scales [23] [24].
- **Multi-Criteria Decision Analysis (MCDM):** Integrates spatial analysis and holistic planning, especially in developing nations [25].

Synthesis: Multi-hazard risk assessment is foundational to BBB, enabling informed land-use and infrastructure planning that anticipates cascading failures and prioritizes resilience investments.

Enforcement of Building Codes and Structural Improvements

- **Risk-Informed Codes:** Multi-hazard building codes, quality assurance, and skilled builder incentives are critical for structural resilience [26].
- **Community Education:** Programs facilitate compliance, promote safer construction, and overcome barriers such as informal sectors and limited information [27] [28].
- **Legislative Support:** Effective post-disaster legislation enforces risk reduction and efficient recovery [4] [29].

Synthesis: Structural improvements under BBB require not only technical standards but also community education and legislative backing to ensure compliance and long-term resilience.

Community Engagement and Education

- **Participatory Planning:** Community involvement tailors recovery to local needs, enhances compliance, and builds trust [30] [31].
- **Education Programs:** Disaster education integrated into curricula and targeted interventions improve preparedness and resilience [28] [32].
- **Empowerment:** Genuine participation and capacity development are essential for sustainable recovery [6].

Synthesis: Community engagement is a linchpin of BBB, fostering legitimacy, transparency, and adaptive capacity in recovery processes.

Legislative and Institutional Frameworks

- **Decentralized Recovery Authorities:** Expedite decision-making and funding access, though transparency and grassroots involvement need improvement [29].
- **Hazard-Based Land-Use Planning:** Legislative models integrate risk-based zoning and stakeholder engagement [33].
- **Institutional Reforms:** Embedding empowerment and inclusive governance is vital for BBB success [7] [34].

Synthesis: Legislative and institutional frameworks must balance central oversight with local autonomy, ensuring risk reduction and community empowerment.

3. Innovative Methodologies and Technologies for Resilient Infrastructure

Dynamic Risk Modeling and Cascading Failure Analysis

- **DBN Optimization:** Incorporating multi-temporal data, Granger causality, and recovery dynamics improves modeling of cascading failures [35] [36] [37].
- **Probabilistic Frameworks:** Efficiently model large-scale interdependent systems, reducing complexity and enabling exact inference [38].
- **Multi-State Node Models:** Enhance robustness and reduce failure propagation compared to binary-state models [39].

Synthesis: Advanced probabilistic models are essential for understanding and mitigating cascading failures in multi-hazard infrastructure systems.

Geospatial Integration and Multi-Hazard Mapping

- **GIS and Remote Sensing:** Fuse multi-source datasets for composite risk indices at fine spatial resolutions [3].
- **Machine Learning Integration:** Predict and map multi-hazard risk zones, supporting urban disaster preparedness [11].
- **Web-Based Platforms:** Improve situational awareness and coordinated decision-making in crisis management [40].

Synthesis: Geospatial integration techniques, increasingly powered by AI, are transforming risk mapping and exposure harmonization in complex urban environments.

Technological Innovations in Risk-Informed Decision-Making

- **AI and IoT:** Real-time monitoring, early warning, and optimized response strategies enhance infrastructure resilience [9] [41].
- **Performance-Based Assessment:** Probabilistic risk analysis and simulation support selection of effective resilience pathways [10].
- **Digital Platforms:** Facilitate community feedback loops and participatory governance [42].

Synthesis: Technological innovations are driving risk-informed, accountable, and adaptive infrastructure decisions, bridging quantitative metrics with community insights.

4. Infrastructure Recovery Strategies and Sustainable Frameworks

Sustainable Recovery and Social Capital Integration

- **Social Capital and Leadership:** Bottom-up participatory approaches empower local governments and civil society, improving recovery outcomes [12] [43].
- **Multi-Objective Optimization:** Balances recovery efficiency and equity, reducing disparities in service restoration [44].
- **Community Learning Centers:** Leverage social networks and local leadership for sustained resilience [45].

Synthesis: Integrating social capital and leadership into recovery frameworks accelerates recovery and enhances equity, especially among vulnerable groups.

Decentralized Recovery Authorities and Institutional Coordination

- **Supervisory Cooperative Strategies:** Integrate local planning with cross-system coordination, reducing recovery time and improving resilience [13].
- **Resource Allocation:** Critical for effectiveness and timeliness; misallocations degrade outcomes [46] [47].
- **Political Interference:** Prioritizing short-term crisis management can bias resource distribution [48].

Synthesis: Decentralized authorities must balance speed and quality, with cross-departmental coordination and administrative interventions optimizing recovery outcomes.

People Centered Housing Recovery (PCHR) in Diverse Contexts

- **Adaptable Frameworks:** Sensitive to local contexts, empowering residents and integrating psychosocial and cultural dimensions [14] [49].
- **Inclusive Engagement:** Meaningful participation tailored to socio-political contexts improves satisfaction and cohesion [50].

- **Owner-Driven Reconstruction:** Enhances resilience through community trust and culturally relevant modifications [51].

Synthesis: PCHR models must be flexible, culturally sensitive, and participatory to address diverse community needs and support long-term resilience.

5. Policy Frameworks for Resilient Infrastructure

Legislative Models and Post-Disaster Policy Effectiveness

- **High-Income vs. Low-Income Contexts:** Resource and capacity constraints hinder reconstruction in low-income countries; high-income contexts better sustain BBB outcomes [5] [52].
- **Disaster-Specific Legislation:** Updating procurement laws and contracts improves reconstruction effectiveness [17].
- **Inclusive Planning:** Participatory processes enhance recovery outcomes and address social vulnerabilities [53].

Synthesis: Effective policy frameworks require continuous alignment of governance, institutional capacity, and community engagement, with tailored approaches for diverse economic contexts.

Institutional Coordination and Local Government Roles

- **Decentralized Mandates:** Enable faster decision-making and funding access, with local councils critical for land-use planning and community consultations [29] [33].
- **Bridging Policy and Practice:** Embedding local knowledge and scientific data into place-based approaches aligns resilience principles with sustainable urbanization [54].
- **Institutional Reforms:** Fiscal decentralization and innovative governance models embed community empowerment [7].

Synthesis: Local governments are pivotal in bridging policy and practice gaps, requiring empowerment and capacity building to ensure effective BBB implementation.

Community Engagement in Policy Implementation

- **Transparency and Legitimacy:** Participatory approaches improve fairness, inclusiveness, and project outcomes [7] [55].
- **Resource Constraints:** Practical implementation often falls short, necessitating a shift to community-oriented recovery support [56].
- **Digital Platforms:** Facilitate coordination and adaptive capacity development [57].

Synthesis: Community engagement is essential for effective policy implementation, enhancing legitimacy, transparency, and adaptive capacity in BBB recovery programs.

6. Community Involvement and Social Infrastructure

Community-Based Interventions and Vulnerable Populations

- **Targeted Interventions:** Comprehensive programs improve disaster resilience among older adults, with results expected in ongoing studies [58].
- **Health Worker Training:** Enhances capacity to promote preparedness and resilience [59].
- **Social Participation:** Strong sense of community belonging is associated with greater resilience [60].

Synthesis: Community-based interventions, especially for vulnerable populations, are critical for improving preparedness, resilience, and recovery outcomes.

Community Learning Centers and Local Knowledge Integration

- **Kominkan Model:** Embeds disaster preparedness into cultural activities, leveraging local leadership and social networks [45].
- **Participatory Knowledge Production:** University-community partnerships strengthen adaptive capacity and resilience [61].
- **Cultural Practices:** Inclusive decision-making and social cohesion enhance infrastructure sustainability [62].

Synthesis: Community learning centers are multifunctional hubs for education, cultural continuity, and disaster readiness, sustaining long-term infrastructure resilience.

Social Capital and Adaptive Capacity

- **Bonding, Bridging, Linking:** Synergistically support disaster preparation, survival, and recovery [63] [64].
- **Equity-Informed Strategies:** Reduce disparities in service restoration times, benefiting peripheral communities [44].
- **Social Infrastructure:** Facilitates access to aid and accelerates recovery [65].

Synthesis: Social capital in its various forms is a powerful determinant of recovery speed, equity, and adaptive capacity in interdependent infrastructure systems.

7. Power Dynamics, Legitimacy, and Governance in Recovery

Power Dynamics and Participation Outcomes

- **Critical Consciousness:** Participatory processes can shift power relations and foster collaborative governance [8].
- **Empowerment and Legitimization:** Directly affect participation outcomes and project success [55].
- **Trust and Social Justice:** Recovery strategies must address institutional trust deficits and rights-based service delivery [66].

Synthesis: Power dynamics and legitimacy are central to meaningful community participation, requiring explicit analysis and institutional reforms.

Institutional Reforms and Fiscal Decentralization

- **Bottom-Up Approaches:** Participatory budgeting and integrated community-based risk reduction enhance empowerment [15].
- **Capacity Building:** Adequate local capacity and clear responsibilities are essential for effective decentralization [67].
- **Justice-Centered Frameworks:** Integrate social inclusion and adaptive governance for resilience building [68].

Synthesis: Institutional reforms and fiscal decentralization strategies embed community empowerment, improving equity and sustainability in recovery cycles.

Transparency, Grassroots Involvement, and Trust Building

- **Community Data Generation:** Empowers marginalized groups and promotes equitable governance [69].
- **Social Entrepreneurs:** Facilitate co-production and community engagement [70].
- **Early Engagement:** Equitable participation and capacity enhancement are critical for sustainable recovery outcomes [71].

Synthesis: Decentralized recovery authorities must prioritize transparency, grassroots involvement, and trust-building to strengthen community resilience post-disaster.

8. Conclusion and Future Directions

Synthesis of Lessons Learned

The Build Back Better framework is a multidisciplinary paradigm where technological innovations, robust legislative frameworks, and meaningful community involvement converge. Advanced risk assessment methodologies and integrated GIS models are paving the way for data-driven recovery strategies, while challenges related to power imbalances and inconsistent community engagement persist. Addressing these gaps through interdisciplinary studies and institutional reforms will further enhance the effectiveness and inclusiveness of disaster recovery efforts [8] [19] [20] [21].

Research Gaps and Recommendations

- **Quantitative Evaluation:** Long-term impact of community education and participatory recovery schemes on compliance with risk-informed building codes [16].
- **Power Dynamics Integration:** Empirical studies on the role of power and legitimacy in recovery models [18].
- **Scalability of Decentralized Models:** Comparative research across cultural and economic contexts [17].
- **Technological-Governance Linkages:** Interdisciplinary studies connecting technological innovations with ground-level recovery outcomes [44].

Creative Insight: Develop hybrid frameworks combining real-time data integration from IoT sensors and remote sensing with active community feedback loops, ensuring both quantitative risk metrics and local socio-cultural insights shape recovery policies.

In summary, resilient infrastructure planning post-disaster under the Build Back Better framework requires the integration of advanced risk assessment methodologies, supportive legislative and policy frameworks, genuine community engagement, and technological innovation. Addressing persistent challenges and research gaps will enable more effective, equitable, and sustainable recovery outcomes globally.

References

1. Toward Temporal Multi-Hazard Risk Assessment Using Dynamic Bayesian Network Analysis Bakhtiari, S., Najafi, M.R., Goda, K., Peerhossaini, H. Lecture Notes in Civil Engineering, 2025 https://www.scopus.com/pages/publications/105023373579?origin=scopusAI

2. A Data-Driven Probabilistic Framework for Cascading Urban Risk Analysis Using Bayesian Networks Lankalapalli, S.V., Kumar, C.R., Shanmuk, P.H.D.S., (...), Dwibedy, D. Lecture Notes in Networks and Systems, 2026 https://www.scopus.com/pages/publications/105028356785?origin=scopusAI

3. Multi-Source Geospatial Analysis for Disaster Risk Management in Smart Cities: Integration of GIS & Remote Sensing Uckardesler, S., Yomralioglu, T. International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives, 2025 https://www.scopus.com/pages/publications/105030488770?origin=scopusAI

4. Legislation for building back better of horizontal infrastructure Bassett, M., Wilkinson, S., Mannakkara, S. Disaster Prevention and Management, 2017 https://www.scopus.com/pages/publications/85009205130?origin=scopusAI

5. Resources and capacity: Lessons learned from post-disaster reconstruction resourcing in Indonesia, China and Australia Chang, Y., Wilkinson, S., Potangaroa, R., Seville, E. COBRA 2010 - Construction, Building and Real Estate Research Conference of the Royal Institution of Chartered Surveyors, 2010 https://www.scopus.com/pages/publications/84860470637?origin=scopusAI

6. Critical success factors for community-based post-disaster housing reconstruction projects in the pre-construction stage in Indonesia Ophiyandri, T., Amaratunga, D., Pathirage, C., Keraminiyage, K. International Journal of Disaster Resilience in the Built Environment, 2013 https://www.scopus.com/pages/publications/84879373609?origin=scopusAI

7. The Challenges and Opportunities of Community Participation in Disaster Recovery Moore, K., Goodstadt, V. Built Environment, 2026 https://www.scopus.com/pages/publications/105036021676?origin=scopusAI

8. Shifting power through participation in post-disaster recovery: A scoping review Tuhkanen, H. International Journal of Disaster Risk Reduction, 2023 https://www.scopus.com/pages/publications/85173404056?origin=scopusAI

9. AI Solutions for Energy Geotechnics and Disaster Management Resilience Kumar, N. Lecture Notes in Civil Engineering, 2025 https://www.scopus.com/pages/publications/85211594081?origin=scopusAI

10. Performance-based selection of pathways for enhancing built infrastructure resilience Joo, M.R., Sinha, R. Sustainable and Resilient Infrastructure, 2023 https://www.scopus.com/pages/publications/85150608762?origin=scopusAI

11. Multi hazard risk assessment through geospatial analysis and machine learning Pidurkar, P., Khedgaonkar, R.S., Mallu, S.R., (...), Badhiye, S.S. Discover Global Society, 2026 https://www.scopus.com/pages/publications/105047014579?origin=scopusAI

12. Social capital and the cultural contexts of disaster recovery outcomes in myanmar and taiwan James, H., Paton, D. Global Change, Peace and Security, 2015 https://www.scopus.com/pages/publications/84931009891?origin=scopusAI

13. Collaborative decision making for natural disaster recovery of interdependent infrastructure systems Magoua, J.J., Huang, J., Kwak, Y.H., Li, N. Reliability Engineering and System Safety, 2026 https://www.scopus.com/pages/publications/105032720924?origin=scopusAI

14. Towards a policy that supports people-centered housing recovery—learning from housing reconstruction after the Hanshin-Awaji Earthquake in Kobe, Japan Maly, E., Shiozaki, Y. International Journal of Disaster Risk Science, 2012 https://www.scopus.com/pages/publications/84907808730?origin=scopusAI

15. Sectoral Integration in Disaster Risk Reduction (DRR) Investments: The Case of Myanmar and Philippines' Financial Disaster Risk Response Maningo, C.A., Phyu, E.E. Thammasat Review, 2023 https://www.scopus.com/pages/publications/85179702591?origin=scopusAI

16. Opportunities and challenges of compliance to safe building codes: Bangladesh and Nepal Ahmed, I., Gajendran, T., Brewer, G., (...), Sitoula, N. APN Science Bulletin, 2019 https://www.scopus.com/pages/publications/85105746989?origin=scopusAI

17. Contracting challenges in post-disaster reconstruction in developing countries: Evidence from Nepal reconstruction Puri, A., Ameer Ali, N., Elkharboutly, M. International Journal of Disaster Risk Reduction, 2025 https://www.scopus.com/pages/publications/105001137892?origin=scopusAI

18. Digging deeper: Participation and non-participation in post-disaster community recovery Chandrasekhar, D. Community Development, 2012 https://www.scopus.com/pages/publications/84871264203?origin=scopusAI

19. Missing the Window: Barriers to Building Back Better in the Western Cape, South Africa Pharoah, R. Natural Hazards and Public Management: Governing Climate Adaptation, 2026 https://www.scopus.com/pages/publications/105031036286?origin=scopusAI

20. Build back better: A framework for sustainable recovery assessment Zhou, B., Zhang, H., Evans, R. International Journal of Disaster Risk Reduction, 2022 https://www.scopus.com/pages/publications/85130334183?origin=scopusAI

21. Building Back Better in the Cook Islands: A Focus on the Tourism Sector Mannakkara, S., Wilkinson, S., Willie, M., Heather, R. Procedia Engineering, 2018 https://www.scopus.com/pages/publications/85043383956?origin=scopusAI

22. Multi-natural hazard mapping for critical infrastructures in complex territorial contexts: proposal of a novel methodological approach Castiglione, M., Capodici, M., Corsino, S.F., (...), Torregrossa, M. Journal of Environmental Management, 2026 https://www.scopus.com/pages/publications/105028486535?origin=scopusAI

23. AN INDEX-BASED METHODOLOGY FOR MULTI-HAZARD RISK ASSESSMENT OF BRIDGE PORTFOLIOS Grieco, L.A., Scattareggia, N., Monteiro, R., Parisi, F. World Conference on Earthquake Engineering proceedings, 2024 https://www.scopus.com/pages/publications/105027847316?origin=scopusAI

24. MODELING OF MULTI-HAZARD DISASTER RISK AND INFRASTRUCTURE RESILIENCE AT THE GLOBAL LEVEL Cardona, O.D., Bernal, G., Marulanda, M., (...), Gonzalez, D. World Conference on Earthquake Engineering proceedings, 2024 https://www.scopus.com/pages/publications/105027922901?origin=scopusAI

25. Multi-hazard Risk Unveiled: Pioneering Techniques for Comprehensive Risk Analysis and Mitigation Mahato, P., Srivastava, S., Jogi, S., Pandey, S. Operations Research Forum, 2025 https://www.scopus.com/pages/publications/85218144274?origin=scopusAI

26. Build Back Better principles for post-disaster structural improvements Mannakkara, S., Wilkinson, S. Structural Survey, 2013 https://www.scopus.com/pages/publications/84883063118?origin=scopusAI

27. Planning for post-disaster recovery and reconstruction Schwab, J., Topping, K.C., Eadie, C.C., (...), Rosenberg, C.G. APA Planning Advisory Service Reports, 1998 https://www.scopus.com/pages/publications/0032317946?origin=scopusAI

28. The implementation of disaster education after the sixteen years of the 2004 Indian Ocean Tsunami in Aceh-Indonesia: Progress or regress? Ridha, S., Rahman, A., Abdi, A.W., Kamil, P.A. E3S Web of Conferences, 2022 https://www.scopus.com/pages/publications/85146831493?origin=scopusAI

29. Selecting an institutional mechanism for Building Back Better: Lessons from Victorian bushfires recovery Mannakkara, S., Wilkinson, S. International Journal of Disaster Risk Reduction, 2016 https://www.scopus.com/pages/publications/84991220660?origin=scopusAI

30. Post-disaster reconstruction and change: Communities' perspectives Barenstein, J.E.D., Leemann, E. Post-Disaster Reconstruction and Change: Communities' Perspectives, 2012 https://www.scopus.com/pages/publications/85061954924?origin=scopusAI

31. Roles of people, community and planning in recovery after Mega-Disasters: A symposium synopsis Iuchi, K., Maly, E. Journal of Disaster Research, 2016 https://www.scopus.com/pages/publications/84973545269?origin=scopusAI

32. Using community education interventions to build resilience and avert crises: how accidental dwelling fires decreased in Essex County, UK Reinhardt, G.Y., Chatsiou, K. Local Government Studies, 2019 https://www.scopus.com/pages/publications/85061082722?origin=scopusAI

33. Build back better: Implementation in Victorian bushfire reconstruction Mannakkara, S., Wilkinson, S., Potangaroa, R. Disasters, 2014 https://www.scopus.com/pages/publications/84895761513?origin=scopusAI

34. Structural and operational factors as determinant of meaningful community participation in sustainable disaster recovery programs: The case of Bangladesh Islam, E., Wahab, H.A., Benson, O.G. International Journal of Disaster Risk Reduction, 2020 https://www.scopus.com/pages/publications/85087954533?origin=scopusAI

35. Multi-temporal damage assessment of linear infrastructural objects using Dynamic Bayesian Networks Frey, D., Butenuth, M. 2011 6th International Workshop on the Analysis of Multi-Temporal Remote Sensing Images, Multi-Temp 2011 - Proceedings, 2011 https://www.scopus.com/pages/publications/80053093121?origin=scopusAI

36. Risk-response coupling in underground structures under liquefiable soil conditions: A causality-informed Dynamic Bayesian network integrated framework Kong, H., Bao, X., Shen, J., (...), Chen, X. Engineering

Applications of Artificial Intelligence, 2025 https://www.scopus.com/pages/publications/105015300199?origin=scopusAI

37. Modeling Interdependencies and Cascading Failures in Critical Infrastructure Using a Multi-System Simulation Framework Nickdoost, N. Construction Research Congress 2026: Advanced Technologies, Artificial Intelligence, and Data Analytics in Construction - Selected papers from Construction Research Congress 2026, 2026 https://www.scopus.com/pages/publications/105049182535?origin=scopusAI

38. Framework for Probabilistic Vulnerability Analysis of Interdependent Infrastructure Systems Applegate, C.J., Tien, I. Journal of Computing in Civil Engineering, 2019 https://www.scopus.com/pages/publications/85056118674?origin=scopusAI

39. Enhancing communication network robustness under adaptive load-capacity model and multistate stochastic Markov cascading failure model Tian, W., Xu, J., Ma, C., Zou, X. Physica A: Statistical Mechanics and its Applications, 2025 https://www.scopus.com/pages/publications/105009854469?origin=scopusAI

40. RIGID: A Workspace-Based Geospatial System for Multi-Hazard Crisis Coordination Tassopoulos, D., Lavasa, A., Konstantinidis, I., (...), Karakostas, A. Proceedings of the International ISCRAM Conference, 2026 https://www.scopus.com/pages/publications/105042483299?origin=scopusAI

41. Disaster risk reduction and management with emerging technologies: Applications of IoT, AI, and data analytics for resilient urban infrastructure Pathak, S., Abbas, U., Abass, S., (...), Chouksey, A. Revolutionizing Urban Development and Governance With Emerging Technologies, 2025 https://www.scopus.com/pages/publications/105004865455?origin=scopusAI

42. AI-enabled resilience modelling across disaster, response, recovery and empirical study of cyber-physical infrastructure with industrial IoT Ardebili, A.A., Padoano, E., Boscolo, M. International Journal of Disaster Risk Reduction, 2026 https://www.scopus.com/pages/publications/105048834745?origin=scopusAI

43. Spatiotemporal Modeling of Social Capital and Infrastructure Mobility Resilience during Hurricane Harvey Ding, R., Li, W., Lei, Z., (...), Wang, Q.R. Natural Hazards Review, 2026 https://www.scopus.com/pages/publications/105041439828?origin=scopusAI

44. Integrating Social Equity into Resilience-based Post-disaster Recovery of Interdependent Infrastructure Systems Huang, J., Liu, M., Magoua, J.J., Li, N. Reliability Engineering and System Safety, 2026 https://www.scopus.com/pages/publications/105039279343?origin=scopusAI

45. The role of social infrastructure in community-based disaster resilience: A case study of the 2024 Noto Peninsula earthquake Ishiwatari, M., Nomura, S., Kanbara, S., (...), Aldrich, D. International Journal of Disaster Risk Reduction, 2026 https://www.scopus.com/pages/publications/105028993924?origin=scopusAI

46. Post-disaster reconstruction of transportation infrastructures: Lessons learned Rouhanizadeh, B., Kermanshachi, S. Sustainable Cities and Society, 2020 https://www.scopus.com/pages/publications/85091567885?origin=scopusAI

47. Post-disaster recovery planning for infrastructure systems based on residents' needs: A hypernetwork approach Zhao, Z., Li, Z., Wang, T., (...), Fang, D. International Journal of Disaster Risk Reduction, 2025 https://www.scopus.com/pages/publications/85216481836?origin=scopusAI

48. Administrative planning and political response to a post-disaster reconstruction: A study of Aila (cyclone)-devastated Gosaba block in West Bengal (India) Bhattacharyya, R. Strategic Disaster Risk Management in Asia, 2015 https://www.scopus.com/pages/publications/84943163026?origin=scopusAI

49. Rebuilding Lives, Not Just Homes: A Capabilities Approach to Post-Disaster Housing Recovery Puriya, A., Biswas, A. Journal of Human Development and Capabilities, 2026

https://www.scopus.com/pages/publications/105044115055?origin=scopusAI

50. Localised actor roles in post-disaster housing recovery: A case study from Kerala Nair, M.N., Baiju, K.R., Mani, S.K., Uppaluri, S. E3S Web of Conferences, 2025
https://www.scopus.com/pages/publications/105017132223?origin=scopusAI

51. A novel framework for owner driven reconstruction projects to enhance disaster resilience in the long term Vahanvati, M. Disaster Prevention and Management: An International Journal, 2018
https://www.scopus.com/pages/publications/85049520311?origin=scopusAI

52. Diagnosing long-term divergence in Build Back Better outcomes in low- and lower-middle-income communities Opabola, E.A., Baiguera, M., Raby, A., (...), Marafini, F. International Journal of Disaster Risk Reduction, 2026 https://www.scopus.com/pages/publications/105039208981?origin=scopusAI

53. Urban planning challenges and missed opportunities after an industrial disaster in Hungary Csizmady, A., Ferencz, Z.J., Bagyura, M. Disaster Prevention and Management, 2026
https://www.scopus.com/pages/publications/105030514685?origin=scopusAI

54. Place Based Approach to plan for Resilient Cities: A local government perspective Mirti Chand, A.V. Procedia Engineering, 2018 https://www.scopus.com/pages/publications/85043385483?origin=scopusAI

55. Empowerment and legitimization of effected communities in post-disaster reconstruction Shafique, K., Warren, C.M.J. Procedia Engineering, 2018 https://www.scopus.com/pages/publications/85043355260?origin=scopusAI

56. The social infrastructure of disaster recovery: policy implications for ‘Build Back Better’ after Cyclone Idai in Malawi Livne, S., Kanyema, V., Nhlema, B., (...), Rosenthal, A. Climate and Development, 2026
https://www.scopus.com/pages/publications/105049149810?origin=scopusAI

57. OPTIMISING COMMUNITY PARTICIPATION IN POST-DISASTER CRISIS RESPONSE: EVIDENCE FROM THE “DITWAH” CYCLONE, SRI LANKA Jayalath, J.A.C.D., Kulatunga, U. World Construction Symposium, 2026 https://www.scopus.com/pages/publications/105046425948?origin=scopusAI

58. Disaster Preparedness Intervention for Older Adults (Seniors’ Positive Involvement in Community Emergencies): Protocol for a Quasi-Experimental Study White-Lewis, S., Lightner, J., Crowely, J., (...), Chesnut, S. JMIR Research Protocols, 2024 https://www.scopus.com/pages/publications/85211354496?origin=scopusAI

59. Training Community Health Workers to Enhance Disaster Resilience Nicholls, K., Picou, S.J., McCord, S.C. Journal of Public Health Management and Practice, 2017
https://www.scopus.com/pages/publications/85030128011?origin=scopusAI

60. Associations Between Resilience, Community Belonging, and Social Participation Among Community-Dwelling Older Adults: Results From the Eastern Townships Population Health Survey Levasseur, M., Roy, M., Michallet, B., (...), Généreux, M. Archives of Physical Medicine and Rehabilitation, 2017
https://www.scopus.com/pages/publications/85030182768?origin=scopusAI

61. EDUCATION FOR DISASTER RISK REDUCTION IN THE SEISMIC LANDSCAPE: CO-CREATING RESILIENCE THROUGH UNIVERSITY-SCHOOL NETWORKS IN NORTHERN THAILAND Juwitasari, R., Inpin, W., Miyake, Y., Dania, M. Humanities, Arts and Social Sciences Studies, 2026
https://www.scopus.com/pages/publications/105037498836?origin=scopusAI

62. COMMUNITY PARTICIPATION AND SOCIAL RECONSTRUCTION POST NATURAL DISASTERS BASED ON LOCAL WISDOM IN SIGI DISTRICT CENTRAL SULAWESI PROVINCE Muhamad, H., Abdulah, D.C., Ariani, N., Afdal Jurnal Ilmiah Ilmu Terapan Universitas Jambi, 2025
https://www.scopus.com/pages/publications/105008783388?origin=scopusAI

63. Relying on social capital in all its forms: Disaster resilience in the Bahamas after Hurricane Dorian Storr, V.H., Craig, A.W., Storr, N.M. International Journal of Disaster Risk Reduction, 2025 https://www.scopus.com/pages/publications/105009229845?origin=scopusAI

64. Social support from bonding and bridging relationships in disaster recovery: Findings from a slow-onset disaster Nguyen-Trung, K., Forbes-Mewett, H., Arunachalam, D. International Journal of Disaster Risk Reduction, 2020 https://www.scopus.com/pages/publications/85078998983?origin=scopusAI

65. Pre-disaster social capital and disaster recovery in wenchuan earthquake-stricken rural communities Wei, J., Han, Y. Sustainability (Switzerland), 2018 https://www.scopus.com/pages/publications/85048642981?origin=scopusAI

66. Community Participation and Power Relations in Post-Disaster Recovery: A Qualitative Study from an Anti-Oppressive Social Work Perspective Erdurak, Y. Journal of Evidence-Based Social Work (United States), 2026 https://www.scopus.com/pages/publications/105046443790?origin=scopusAI

67. How does decentralisation affect health sector planning and financial management? a case study of early effects of devolution in Kilifi County, Kenya Lucy Gilson Tsofa, B., Molyneux, S., Gilson, L., Goodman, C. International Journal for Equity in Health, 2017 https://www.scopus.com/pages/publications/85029359266?origin=scopusAI

68. Regeneration beyond growth: Reimagining urban futures through justice and climate resilience Adamu, S., Yong, H., Alhaji, M., (...), Ibrahim, M.A. Habitat International, 2026 https://www.scopus.com/pages/publications/105035669436?origin=scopusAI

69. Enhancing equity of the post-disaster recovery governance through community data generation Rodríguez-Gaviria, E.M., Rivera-Flórez, L.A., Albuquerque, J.P.D. International Journal of Disaster Risk Reduction, 2024 https://www.scopus.com/pages/publications/85200799540?origin=scopusAI

70. Social entrepreneurship, co-production, and post-disaster recovery Rayamajhee, V., Storr, V.H., Bohara, A.K. Disasters, 2022 https://www.scopus.com/pages/publications/85092553843?origin=scopusAI

71. Toward participatory participation: A community perspective on effective engagement in post-disaster recovery and reconstruction Ngulube, N.K., Tatano, H., Samaddar, S. Community Development, 2024 https://www.scopus.com/pages/publications/85199294750?origin=scopusAI